

AMSS 2026 — Lecture 12: Presentation & Defense Skills

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Today's Agenda

1. Frame: the defense is the integrity check
2. Presenting an AI-mediated design
3. Live opener: ask AI to defend a design cold
4. The rubric explained: Critique, Rationale, Traceability — strong vs weak
5. What examiners look for; red flags
6. The dry run: Lab 6 this week

Recap: Architect and Critic

- ▶ **Architect / director:** drive AI through the software development lifecycle (SDLC).
- ▶ **Critic / reviewer:** read AI's output and name what's wrong or missing.

For twelve weeks you played both roles. Today: how to **present and defend** the trail you built.

The Defense Is the Integrity Check

- ▶ **3 of 8 project points.** Cold, unaided, no AI in the room.
- ▶ It's where the trail you built pays off — or where its absence shows.

Today's question: **can you read, justify, and trace your design without AI's help?**

The Literacy Floor: Critique, Rationale, Traceability

From Week 1, and now the explicit subject:

- ▶ **Critique** — read & critique any AI-generated artifact on the spot.
- ▶ **Rationale** — articulate why you directed AI a certain way; what you accepted or rejected.
- ▶ **Traceability** — defend the trace: navigate your project end to end.

Today we make each one concrete — what a passing answer sounds like, and what fails.

Present the Trail, Not the Artifacts

Your value was **direction and correction** — so present that, not just the final diagram.

- ▶ Show a prompt that **failed**, and how you fixed it.
- ▶ Show an output you **rejected**, and why.
- ▶ Show a defect you **caught**, and the correction.

The narrative is the star; the polished diagram is just its result.

Structure of a Good Slice Presentation

A repeatable 5-minute shape:

1. The **requirement** your slice realises.
2. How you **directed** AI to model it.
3. What AI **produced** — and what you **caught**.
4. The **correction** you made and why.
5. The **trace**: requirement to test.

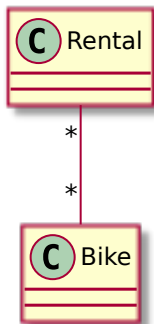
Demo: Ask AI to Defend a Design

Live: paste a design and ask AI to defend it as if in the oral exam. Watch it invent a rationale it never had — and flatter.

Prompt to AI: *“Defend this bike-sharing design as if in an oral exam: why these choices, and is it correct? [paste a design]”*

Critique — Read & Critique Cold

The examiner shows you **any** AI-generated diagram — possibly from another team's repo — and asks: *“What’s wrong with this?”*



You must find the defect on the spot, with no preparation.

Your Turn: Cold Read

What's wrong with this diagram? Read the multiplicity aloud — discuss with your neighbour for 60 seconds — before we compare strong vs weak.

Critique: Strong vs Weak

- ▶ **Weak:** *"It looks fine to me." / "The AI made it, so it should be okay."*
- ▶ **Strong:** *"Read the multiplicity aloud — many-to-many. A rental is one bike. That's wrong; it should be Rental * -- 1 Bike."*

The strong answer **names the defect, reads it aloud, gives the fix.**

Rationale — Articulate Rationale

The examiner asks: *“Why did you direct AI this way? What did you reject?”*

This is where students who let AI drive everything fall silent — they have no rationale because they made no decisions.

Rationale: Strong vs Weak

- ▶ **Weak:** *“The AI suggested it.” / “That’s how the tutorial did it.”*
- ▶ **Strong:** *“I named the domain rule ‘a rental is one bike’ because the many-to-many hid that you borrow a specific bike. I rejected the `BikeShareSystem` god class because responsibilities belonged on `Rental` and `Station`.”*

The strong answer ties each choice to a **reason**, not an authority.

Traceability — Defend Traceability

The examiner asks: *“Show me the test for this requirement”* or *“Which requirement needs this class?”*



You must walk the chain, either direction, on demand.

Traceability: Strong vs Weak

- ▶ **Weak:** *"I think there's a test somewhere..."* — can't connect the layers.
- ▶ **Strong:** *"REQ-1 (per-minute fare) is realised by the Rent Bike use case, the Rental class with a FareStrategy, and verified by the 60-minute fare test. I found and fixed a stale link where the test still checked a flat fare."*

The strong answer **walks the chain and names a link they repaired.**

What Examiners Look For

- ▶ **Ownership** — you drove the design; you can say why.
- ▶ **Judgment** — you can read a diagram and defend your choices.
- ▶ **Honesty** — you can name what's *still* weak in your slice.

Not polish. Not “it works.” Not how many patterns you used.

Red Flags That Fail a Defense

- ▶ *“The AI said it was good.”* — you delegated judgement (see Week 11).
- ▶ Can’t read your **own** diagram.
- ▶ Can’t trace a requirement to its test.
- ▶ Every answer is *“AI did it”* with no rationale.

The Dry Run: Lab 6 This Week

Lab 6 is your **cold-defense dry run**:

- ▶ The instructor picks students **at random** and asks Critique / Rationale / Traceability questions.
- ▶ Informal — but every gap surfaced must be fixed before Week 14.

Treat it as the real thing.

How to Prepare

- ▶ Run your own **trace audit** (Week 10) — fix broken links before someone finds them.
- ▶ **Re-read every AI diagram** in your slice — could you critique it cold?
- ▶ Rehearse the “**why**” for each decision.
- ▶ Know your **defect log** cold — it’s your Critique and Rationale evidence.

The Loop, On Yourself

present -> get critique (peer / instructor) -> fix the gaps -> re-present.

Same architect-and-critic loop as all semester — now applied to your own defense.

The Human Defends

AI built the artifacts. Only **you** can defend them cold — read them, justify them, trace them.

That's the entire point of the unaided rule: the defense measures exactly what AI cannot do for you.

Next Week: Workshop + Finals

- ▶ **Week 13** — open workshop / Q&A: supervised work, final adjustments.
- ▶ **Week 14** — final presentations + per-student cold defense (with Lab 7).

Bring your slice to the workshop with its gaps already mapped.

Critique, Rationale, Traceability — The Through-Line

Today you learned to perform the rubric:

- ▶ **Critique** — read and critique cold; name the defect, read it aloud, give the fix.
- ▶ **Rationale** — tie every choice to a reason, not an authority.
- ▶ **Traceability** — walk the trace either direction, and name a link you fixed.

That's It For Today

- ▶ This week's lab (Lab 6): cold-defense dry run — treat it as real.
- ▶ Next: Week 13 workshop, then Week 14 final presentations.

Questions?